

SYED MUHAMMAD MUJTABA ABBAS

Computer Science Undergraduate

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PROFESSIONAL SUMMARY

Computer Science undergraduate at PAF-KIET with hands-on experience gained through academic projects, software engineering implementations, and a frontend development internship. Strong foundation in software engineering, data structures and algorithms, graph theory, databases, computer networks, and information security. Experienced in developing practical software solutions using multiple programming languages and technologies, with a commitment to continuous learning and professional growth.

TECHNICAL SKILLS

Languages: C#, Java, Python, C, PHP, JavaScript

Frameworks & Libraries: React.js, Vite, FastAPI

Web Development: HTML, CSS, JavaScript, WordPress, REST APIs

Mobile Application Development: Java, Android Studio, XML, SharedPreferences

Databases: SQL, MySQL, Microsoft SQL Server, Firebase, Supabase, Relational Database Design

Core CS Concepts: Data Structures & Algorithms, Graph Theory, Object-Oriented Programming (OOP), Software Engineering (SDLC), Database Management Systems (DBMS & ADBMS), Computer Networks, Information Security, Human-Computer Interaction (HCI), Artificial Intelligence (AI)

EXPERIENCE

Website Frontend Developer Intern — KDYS Lab (3 Months)

- Developed and maintained website frontend features using React.js.
- Worked with PHP to support backend integration for web pages.
- Participated in UI component testing, responsiveness tuning, and bug fixing.

PROJECTS

SmartRoute – Route Optimization & DSA Visualization Platform

Technologies: React.js, Vite, Python, FastAPI, Graph Theory, DSA (BFS, DFS, Dijkstra, A), REST APIs, CSS3*

- Built a full-stack route optimization and algorithm visualization web platform demonstrating practical Data Structures and Algorithms.
- Implemented core graph algorithms from scratch (BFS, DFS, Dijkstra, and A* Search) with real-time interactive step-by-step visual exploration.
- Developed a responsive React.js frontend with dual graph support: sample network grid and real Karachi city network featuring 61 interconnected locations.
- Engineered an asynchronous Python FastAPI backend for optimal pathfinding computations and realistic travel time calculations based on road speeds (km/h).
- Integrated algorithm performance comparisons, interactive animation controls, and educational DSA walkthroughs.

Career AI – AI-Powered Career Guidance Platform

Technologies: React.js, Python, FastAPI, Supabase, PostgreSQL, REST APIs, AI/LLM, JavaScript, HTML, CSS

- Developed a full-stack AI-powered career guidance platform using React.js for the frontend and Python/FastAPI for the backend.
- Integrated AI/LLM technology to provide personalized career guidance, skill-based recommendations, and interactive career assistance.
- Integrated Supabase as the database for securely storing and managing user and application data.
- Implemented user authentication and REST API communication between the frontend and backend.
- Built a responsive, modern, and user-friendly interface with properly integrated frontend, backend, AI services, authentication, and database functionality.

AI-Powered Chatbot Web Application

Technologies: React.js, Vite, Python, FastAPI, REST API, AI/LLM API, HTML, CSS, JavaScript

- Developed a modern AI-powered conversational web application with a responsive SaaS-style user interface.
- Built a React.js + Vite frontend for real-time user interaction, chat management, message rendering, and responsive UI/UX.
- Developed a Python FastAPI backend to handle API requests, process user messages, and communicate securely with an AI language model.

- Implemented RESTful API communication between the frontend and backend for seamless message exchange.
- Integrated an AI/LLM API to generate intelligent, context-aware responses to user queries with error handling for API quotas and fallbacks.

AI Agent Platform – Intelligent Multi-Task AI Assistant

Technologies: React.js, Vite, Python, FastAPI, REST APIs, AI/LLM Integration, JavaScript, HTML, CSS

- Developed an AI Agent Platform designed to handle multiple types of user requests through a centralized intelligent interface.
- Built a modern React.js + Vite frontend with a responsive and interactive UI focused on usability and user experience.
- Developed a Python/FastAPI backend responsible for processing requests, executing application logic, and communicating with external APIs and AI services.
- Integrated capabilities to handle natural-language requests, financial queries, calculations, and local system-level workflows.

Car Care Android Application

Android Studio • Java • XML • SharedPreferences

- Developed a native Android application to help users manage vehicle information, fuel records, maintenance, and expenses.
- Implemented Fuel Tracker with automatic cost calculation, Service Checklist, and visual Expense Tracker with local persistence.

Car Racing Game – Web Browser Game

HTML5 Canvas • JavaScript • CSS3 • Game Development

- Built a fully playable responsive arcade car racing game with three difficulty levels, coin collection system, and collision mechanics.
- Implemented 60fps game loop with requestAnimationFrame and dual input support (desktop keyboard arrows and mobile touch controls).

Hostel Management System – Web-Based Management System

Advanced Database Management Systems Project • PHP, MySQL, PDO

- Developed a web-based hostel management system using a 3-Tier Architecture with PHP and MySQL.
- Implemented parameterized queries (PDO) and secure input validation to neutralize SQL injection vulnerabilities.

Shopping Management System – Desktop Application

Database Management Systems Project • C# Windows Forms, Microsoft SQL Server

- Designed and implemented a desktop shopping management system integrated with Microsoft SQL Server for relational data storage.

Hospital Management System – Patient & Appointment Records

Data Structures & Algorithms Project • Python, Google Colab

- Developed an algorithmic hospital management system in Python utilizing core data structures (min-heaps, BSTs) for efficient patient records and triage.

Clothing Brand Network Design & Simulation

Computer Networks Project • Cisco Packet Tracer

- Designed and simulated an enterprise network configuring routers, switches, VLANs, static routing, DHCP, DNS, and FTP.

EDUCATION

Bachelor of Science in Computer Science (BSCS)

PAF-KIET (Karachi Institute of Economics and Technology), Karachi • 2023 – 2027 (In Progress)

Intermediate (Pre-Engineering)

DHACCS Degree College, Karachi

SOFT SKILLS

Strong Problem-Solving, Fast Learner, Team Collaboration, Good Communication, Attention to Detail, Analytical Thinking